Executive Summary: Moving from First-Dispatch to Batched Dispatch

For: Rider Operations leadership, Marketplace Program Manager, Finance partner, Dispatch engineering lead

Basis: Monte Carlo simulation of 1,000 seeded batching windows (14,795 rider requests). All data is synthetic. None of it comes from Uber or any employer.

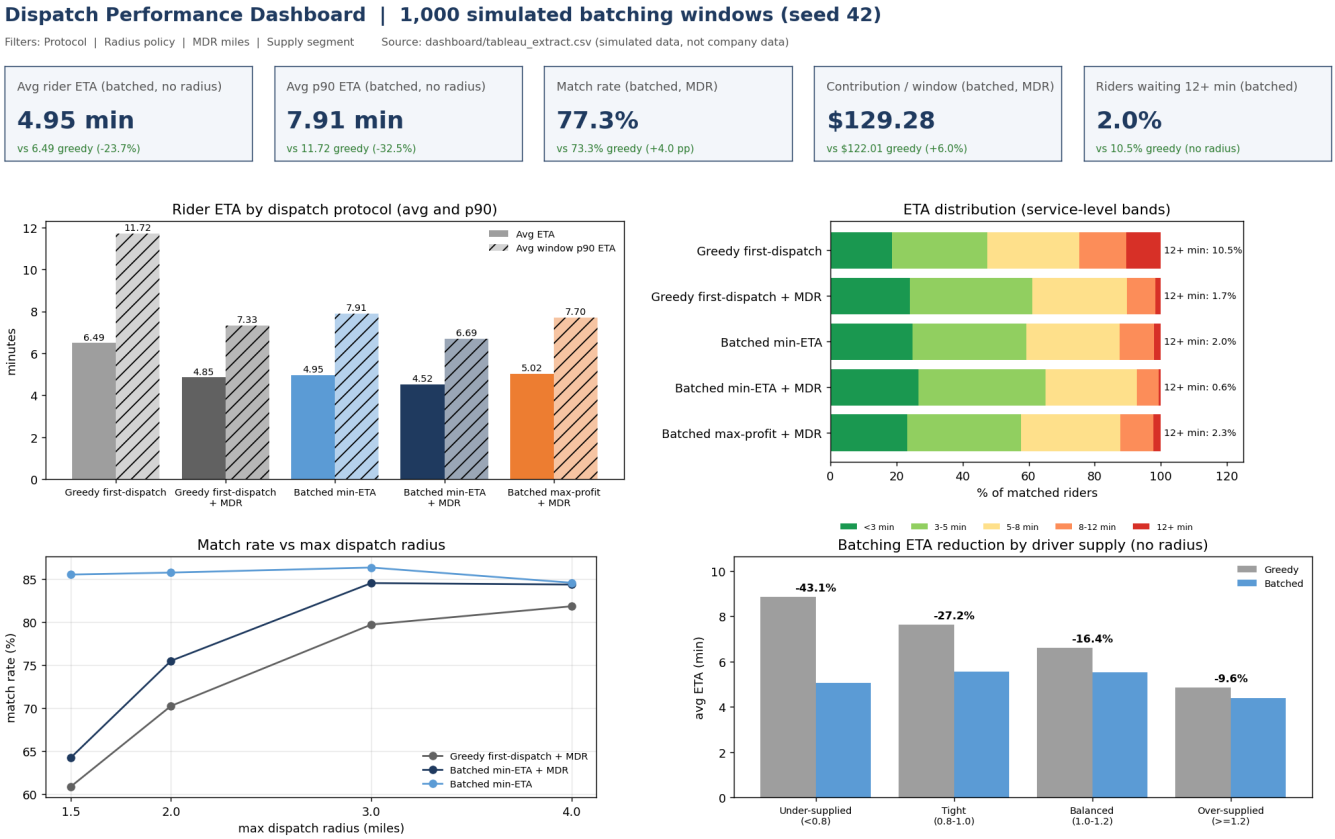
Recommendation

Replace greedy first-dispatch with batched assignment that minimizes total rider ETA, cap the dispatch radius at 3 miles, and roll the change out through a zone-level pilot tracked on the Dispatch Performance Dashboard. In the simulation this cuts average rider ETA by 23.7% with no change in match rate. With the radius in place it also serves 3.98 percentage points more riders and adds 5.96% modelled contribution.

The problem with first-dispatch

- Greedy first-dispatch gives each request the nearest open driver in arrival order. Early requests take drivers that later requests needed, and the tail grows: 10.49% of matched riders wait 12+ minutes, and the average window p90 ETA is 11.72 minutes.
- On the same riders and drivers, greedy dispatch costs 82.12 unpaid en-route minutes per window. Batching costs 62.66.
- Once a dispatch radius is enforced, greedy dispatch strands riders it could have served. Its match rate is 73.33%, against 77.31% for batching.

Results



KPI	Greedy first-dispatch	Batched min-ETA	Change
Avg rider ETA (min), no radius	6.49	4.95	-23.69%
Avg window p90 ETA (min), no radius	11.72	7.91	-32.53%
Riders waiting 12+ min, no radius	10.49%	2.05%	-8.44 pp
Match rate, 1.5-4 mile radius	73.33%	77.31%	+3.98 pp
Avg rider ETA (min), with radius	4.85	4.52	-6.72%
Contribution per window, with radius	$122.01	$129.28	+5.96%

- The gain is largest when drivers are scarce. In under-supplied windows (fewer than 0.8 drivers per rider, 310 windows) batching cuts ETA by 43.06% (3.81 minutes per rider). In over-supplied windows (360 windows) the gain is 9.60% (0.47 minutes).
- The radius is the main trade-off. At 3 miles batching keeps a 84.54% match rate, close to 86.35% with no radius, at 4.89 minutes average ETA. At 1.5 miles the match rate falls to 64.25%.
- Maximizing contribution instead of minimizing ETA adds 12.01% contribution over greedy (radius enforced), but average ETA rises 3.66% versus greedy. That is why it is not the default.

Risks and how the dashboard catches them

The gap analysis flags 506 of 1,000 windows where batching looks worse on at least one KPI. None of them (0) is a true ETA regression on the same riders. In 255 windows average ETA is higher only because batching served extra riders who would otherwise go unmatched. 85 have a longer p90 tail with a better average, and 166 earn less contribution because the objective ignores fare. The dashboard labels each flag by cause, so operations can tell these apart from a real regression before deciding to roll back.

Proposed plan

1. Weeks 1-2: agree KPI definitions and guardrails (avg ETA, p90 ETA, match rate, unmatched riders, share of drivers matched, contribution) with Finance and Dispatch engineering.
2. Weeks 3-6: switch-back pilot in a set of zones, comparing batched min-ETA with a 3-mile radius against current first-dispatch. Review the dashboard daily.
3. Week 7: go or no-go. Go if p90 ETA improves, match rate does not drop, and no zone shows a true regression for 3 consecutive days.
4. After launch: test a contribution-weighted objective, but only in zones where ETA is already within target.

Assumptions and limits

- 6 x 6 mile zone, 18 mph average speed, lognormal traffic noise, about 15 requests per window, design supply ratios of 0.6 to 1.5 drivers per rider, radius drawn from 1.5, 2, 3 and 4 miles.
- Contribution is fare (base $2.50 + $1.25 per trip mile + $0.30 per trip minute) minus $0.65 per unpaid en-route mile. It is a model input, not a P&L figure.
- Unmatched riders do not carry into the next window, and driver repositioning and cancellations are not modelled. These figures need a field pilot before they are used for planning.